

INTERRUPTS

Use of interrupt

Polling

The CPU executes a program that check for the available of data
If a key is pressed then read the data, otherwise keep waiting
or looping!!!

Interrupt

The CPU executes other program, as soon as a key is pressed, the
Keyboard generates an interrupt. The CPU will response to the
interrupt – read the data. After that returns to the original program.
So by proper use of interrupt, the CPU can serve many devices at the
“same time” from the user’s point of view

An Interrupt is either a Hardware generated CALL (externally derived from a hardware signal)

OR

A Software-generated CALL (internally derived from the execution of an instruction or by some other internal event)

- An interrupt is used to cause a temporary halt in the execution of program.
- The meaning of 'interrupts' is to break the sequence of operation.
- While the Microprocessor is executing a program, an 'interrupt' breaks the normal sequence of execution of instructions, diverts its execution to some other program called Interrupt Service Routine (ISR).
- After executing ISR, IRET returns the control back again to the main program. Interrupt processing is an alternative to polling.

IRET instruction

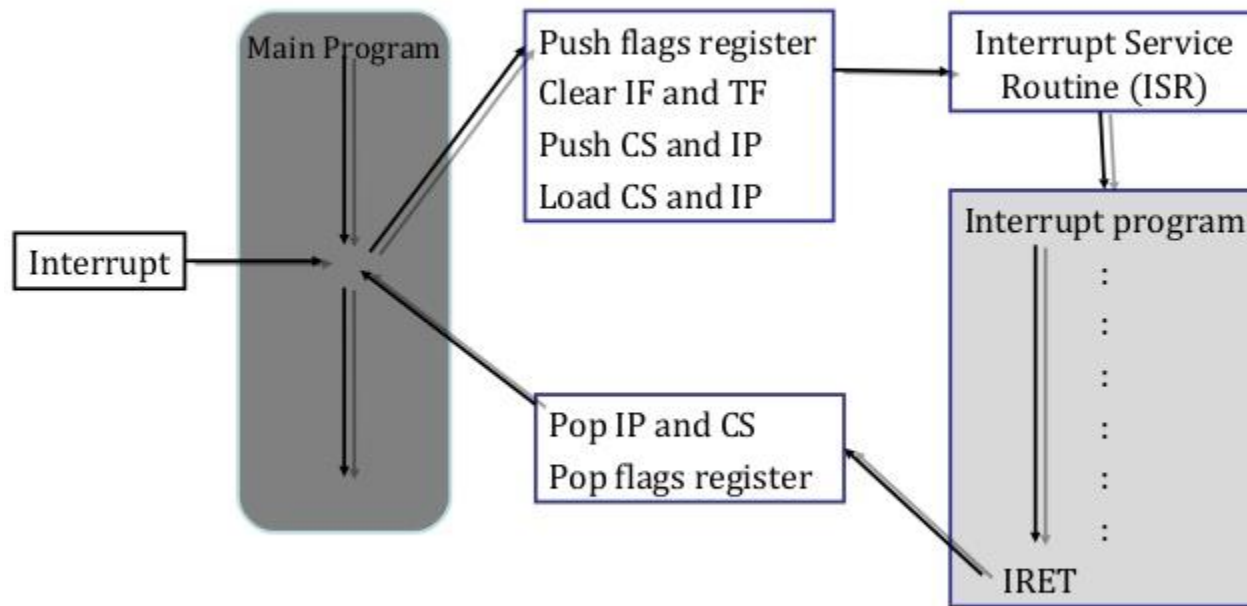
- IRET instruction is used to return from the ISR to the interrupted process and is written at the end of the ISR.
 - $(IP) \leftarrow (SS:SP)$ $SP \leftarrow SP + 2$
 - $(CS) \leftarrow (SS:SP)$ $SP \leftarrow SP + 2$
 - $(Flags) \leftarrow (SS:SP)$ $SP \leftarrow SP + 2$

- INT- Interrupt instruction with type number
- It is 2-byte instruction. First byte provides the op-code and the second byte provides the interrupt type number. There are 256 interrupt types under this group.

Steps to respond an interrupt

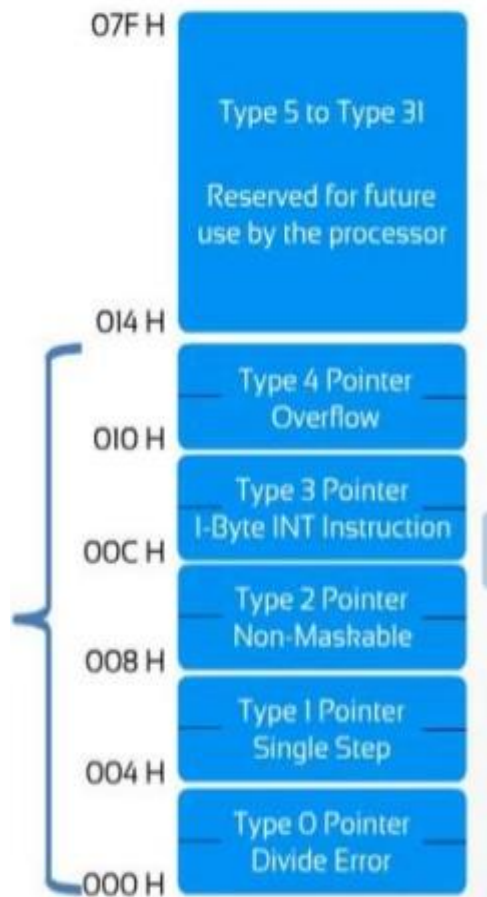
1. Decrement SP by 2 and push the Flag register on to the stack.
2. Clear Interrupt Flag
3. Clear Trap Flag
4. Decrement SP by 2 CS value of the return address pushed on to the stack.
5. Decrement SP by 2 IP value of the return address are pushed on to the stack.
6. Make an indirect far jump to an ISR by loading CS and IP by its address from the vector table.

Processing of an Interrupt by the 8086



- The first five pointers are dedicated interrupt pointers. i.e. –
 - **TYPE 0** interrupt represents division by zero situation.
 - **TYPE 1** interrupt represents single-step execution during the debugging of a program.
 - **TYPE 2** interrupt represents non-maskable NMI interrupt.
 - **TYPE 3** interrupt represents break-point interrupt.
 - **TYPE 4** interrupt represents overflow interrupt.
(INTO-Interrupt if Overflow)

- The interrupts from Type 5 to Type 31 are reserved for other advanced microprocessors,
- Interrupts from 32 to Type 255 are available for hardware and software interrupts

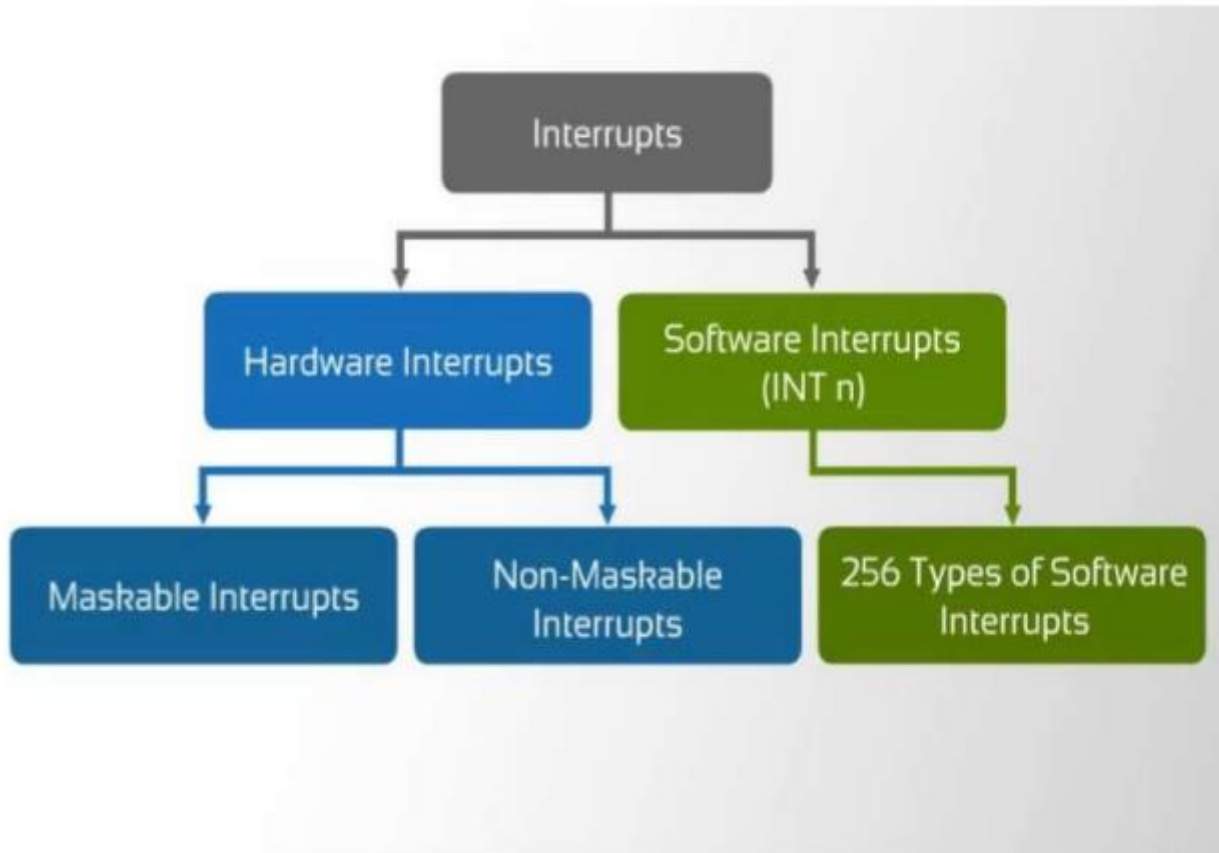


INTERRUPT PRIORITY

- When more than one interrupt occur then priorities of the interrupts are compared in order to determine which interrupt to serve first.
- Priority hierarchy groups: internal interrupt, nonmaskable interrupt, software interrupt, and external hardware interrupt
- Internal interrupt group has the highest priority
- External hardware interrupt group has the lowest priority
- Within a group, different interrupts have different priority levels represented by the type number (or interrupt number)

Interrupt type	Priority level
Divide error, INT n, INTO	Highest
Non-maskable	Very high
INTR	High
Single step	Lowest

Types of Interrupt

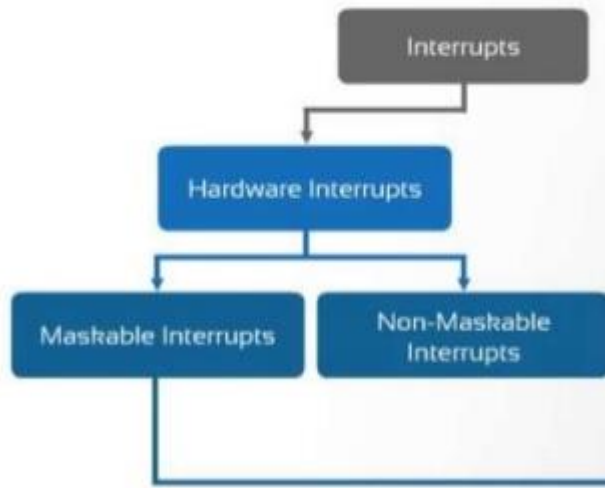


Hardware Interrupts

- Hardware interrupt is caused by any peripheral device by sending a signal through a specified pin to the microprocessor.
- The 8086 has two hardware interrupt pins, i.e. NMI and INTR. NMI is a non-maskable interrupt and INTR is a maskable interrupt having lower priority. One more interrupt pin associated is INTA called interrupt acknowledge.

INTR

- The INTR is a maskable interrupt because the microprocessor will be interrupted only if interrupts are enabled using set interrupt flag instruction. It should not be enabled using clear interrupt Flag instruction.
- The INTR interrupt is activated by an I/O port. If the interrupt is enabled and NMI is disabled, then the microprocessor first completes the current execution and sends '0' on INTA pin twice. The first '0' means INTA informs the external device to get ready and during the second '0' the microprocessor receives the 8 bit, say X, from the programmable interrupt controller.
- **These actions are taken by the microprocessor –**
 - First completes the current instruction.
 - Activates INTA output and receives the interrupt type, say X.
 - Flag register value, CS value of the return address and IP value of the return address are pushed on to the stack.
 - IP value is loaded from the contents of word location $X \times 4$
 - CS is loaded from the contents of the next word location.
 - Interrupt flag and trap flag is reset to 0



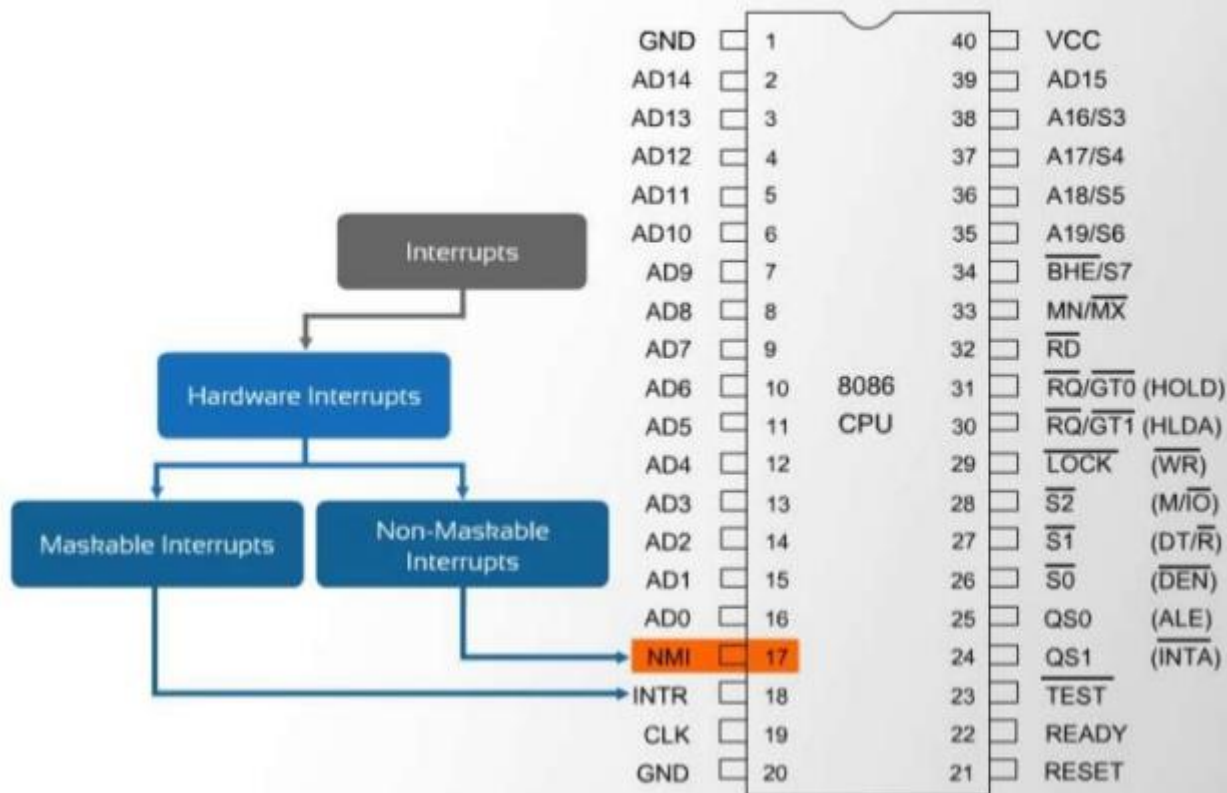
8086 CPU

GND	1	40	VCC
AD14	2	39	AD15
AD13	3	38	A16/S3
AD12	4	37	A17/S4
AD11	5	36	A18/S5
AD10	6	35	A19/S6
AD9	7	34	$\overline{\text{BHE}}/\text{S7}$
AD8	8	33	$\text{MN}/\overline{\text{MX}}$
AD7	9	32	$\overline{\text{RD}}$
AD6	10	31	$\overline{\text{RQ}}/\overline{\text{GT0}}$ (HOLD)
AD5	11	30	$\overline{\text{RQ}}/\overline{\text{GT1}}$ (HLDA)
AD4	12	29	$\overline{\text{LOCK}}$ ($\overline{\text{WR}}$)
AD3	13	28	$\overline{\text{S2}}$ ($\text{M}/\overline{\text{IO}}$)
AD2	14	27	$\overline{\text{S1}}$ ($\text{DT}/\overline{\text{R}}$)
AD1	15	26	$\overline{\text{S0}}$ ($\overline{\text{DEN}}$)
AD0	16	25	QS0 ($\overline{\text{ALE}}$)
NMI	17	24	QS1 ($\overline{\text{INTA}}$)
INTR	18	23	$\overline{\text{TEST}}$
CLK	19	22	READY
GND	20	21	RESET

NMI

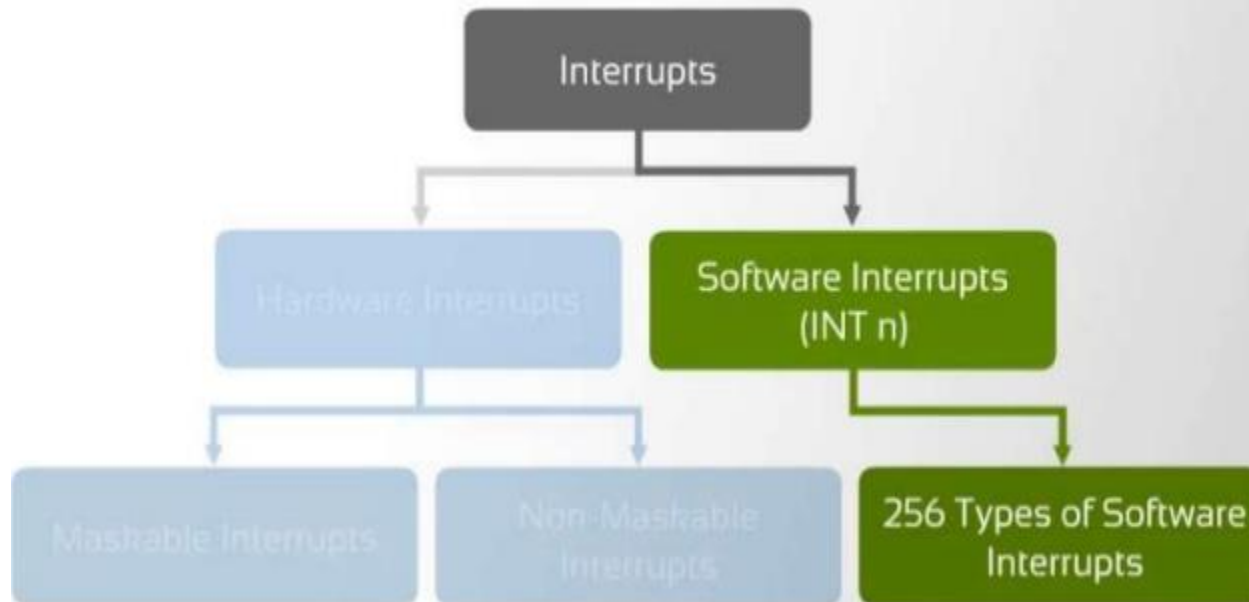
- It is a single non-maskable interrupt pin (NMI) having higher priority than the maskable interrupt request pin (INTR) and it is of type 2 interrupt.
- When this interrupt is activated, these actions take place –
- Completes the current instruction that is in progress.
- Pushes the Flag register values on to the stack.
- Pushes the CS (code segment) value and IP (instruction pointer) value of the return address on to the stack.
- IP is loaded from the contents of the word location 00008H.
- CS is loaded from the contents of the next word location 0000AH.
- Interrupt flag and trap flag are reset to 0.

8086 CPU



Software interrupts

INT stands for Interrupt.



- Software interrupts are like hardware interrupts which are generated by the program itself. From the interrupt number, the CPU derives the address of the Interrupt service routine which must be executed.
- Software interrupts in assembly language can be treated as calls to subroutines of other programs which are currently running on the computer.
- One of the most famous software interrupt is Interrupt No. 21H, which branches in the operating system, and permits the use of PC-DOS functions defined there.
- The function required to be performed by DOS is specified in AH prior to the interrupt.
- The functions return and accept values in various registers. An interrupt is called using the instruction INT followed by the interrupt number . For example: INT 21H

Example ISR

- To display the message “interrupt processed” when an interrupt type 5 occurs.

Data segment

Mes db “interrupt processed”

Data ends

Stack_seg segment stack

DW 20 dup(0)

stack_top label word

Stack ends

Code segment

Assume cs:code,ds:data,ss: stack_seg

Mov AX,data

Mov DS,ax

MOV AX, stack_seg

MOV SS,ax

MOV SP, offset stack_top

MOV AX,0000

MOV ES,AX

MOV word ptr ES:0014h,offset prt_mes

MOV word ptr ES:0016h,seg prt_mes

INT 05

Prt_mes PROC far

PUSH AX

PUSH DX

MOV AH,09

LEA DX,mes

POP DX

POP AX

IRET

Prt_mes ENDP

Code ends

End

Thank you